

Pharmacovigilance Individual Case Processing and Narrative Writing
A Simulated Study Project

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Purpose: Prepared for academic and pharmacovigilance training purposes using simulated data only.

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1. Introduction and Purpose of the Project

Pharmacovigilance plays a critical role in ensuring patient safety through the systematic collection, evaluation, and reporting of adverse drug reactions throughout the product lifecycle. As part of routine pharmacovigilance operations, Safety Associates are required to assess Individual Case Safety Reports (ICSRs), evaluate minimum case validity, document safety-relevant information accurately, and prepare structured case narratives in compliance with established guidelines.

The purpose of this project is to demonstrate foundational pharmacovigilance competencies relevant to an entry-level Safety Associate role by simulating the end-to-end processing of ICSRs within a controlled academic environment. This project was designed to provide hands-on exposure to key operational activities, including case intake, minimum required data assessment, seriousness evaluation, documentation of suspect and concomitant medications, outcome reporting, and action taken with the suspect drug.

A mock dataset comprising 30 Individual Case Safety Reports was created for this project. The dataset intentionally includes a mix of valid and invalid cases to reflect real-world data variability and to demonstrate the application of minimum case validity criteria. Core safety data fields were populated to support structured case review, while incomplete or missing information was retained in selected cases to illustrate appropriate documentation practices for unreported data.

In addition, a subset of five cases was selected for detailed narrative writing to demonstrate the ability to convert structured safety data into clear, medically coherent case narratives. Basic quality checks and a high-level summary analysis were also performed to reflect awareness of data completeness, consistency, and regulatory timelines, without performing clinical interpretation or signal detection activities.

All data used in this project are simulated and were generated solely for academic and training purposes. No real patient information, healthcare professional data, or live safety cases were used. The scope of this project is limited to demonstrating operational understanding of ICSR processing and documentation workflows and does not include regulatory submission, medical review, causality assessment, or benefit–risk evaluation.

This project serves as a practical demonstration of entry-level pharmacovigilance skills required for effective ICSR processing within a Safety Associate role.

2. Project Scope & Objectives

A) Project Scope

The activities performed within the scope of this project include:

- Creation and review of a mock ICSR dataset containing both valid and invalid cases
- Documentation of core safety data fields in a structured format
- Assessment of minimum case validity criteria based on available information
- Classification of seriousness, outcomes, and actions taken with the suspect drug
- Preparation of structured case narratives for a selected subset of cases
- Application of basic quality checks to assess data completeness and internal consistency

Note: The project does not include causality assessment, signal detection, aggregate reporting, regulatory submissions, or benefit–risk evaluation.

B) Project Objectives

- To simulate end-to-end Individual Case Safety Report (ICSR) processing using a structured mock dataset.
- To apply minimum case validity criteria for classification of valid and invalid safety cases.
- To document core safety data fields in accordance with pharmacovigilance documentation standards.
- To prepare structured case narratives for selected cases based on available safety information.
- To perform basic quality and compliance checks to assess data completeness and consistency.

3. Dataset Description

A) Overview of the Dataset

The dataset used in this project consists of 30 simulated Individual Case Safety Reports (ICSRs) created to replicate routine pharmacovigilance case processing activities. The dataset was designed to reflect common real-world scenarios encountered during safety data intake, including both complete and incomplete case information.

Each case represents a unique safety report and is assigned a distinct case identifier to support traceability and documentation throughout the project lifecycle.

B) Case Composition: Valid and Invalid Cases

The dataset includes a mix of valid and invalid cases to demonstrate the application of minimum case validity criteria in pharmacovigilance practice.

- Valid cases contain all four minimum required elements:
 - An identifiable patient
 - An identifiable reporter
 - A suspect drug
 - A reported adverse event
- Invalid cases are those in which one or more of the minimum required elements are missing or not reported at the time of case assessment.

The intentional inclusion of invalid cases allows for practical demonstration of appropriate documentation, classification, and handling of incomplete safety reports in accordance with standard pharmacovigilance workflows.

C) Core Safety Data Fields Included

The dataset captures essential pharmacovigilance data fields commonly used during ICSR processing. These include, but are not limited to:

- Case identifier
- Reporter type
- Patient demographics (age and gender)
- Suspect drug(s)
- Indication for use
- MedDRA PT
- Adverse event(s) reported
- Seriousness classification and seriousness criteria
- Outcome of the adverse event
- Country of occurrence
- Event onset date and case receipt date
- Regulatory due date
- Case validity status and justification

Additional fields such as action taken with the concomitant medications, relevant medical history, Dose, Duration of Therapy and narrative text were included where applicable to support comprehensive documentation.

D) Nature of the Data

All data contained within this dataset are entirely simulated and were generated solely for academic and training purposes. No real patient information, healthcare professional data, or live pharmacovigilance cases were used or referenced.

The dataset does not represent actual safety signals, clinical outcomes, or regulatory submissions. Its purpose is limited to demonstrating operational understanding of ICSR documentation, assessment, and quality considerations expected of an entry-level Safety Associate.

Summary

The dataset comprises 30 simulated ICSRs with a balanced mix of valid and invalid cases, designed to support practical demonstration of core pharmacovigilance documentation and assessment activities.

4. Case Processing Methodology

A) Case Intake

Case intake represents the initial step in the ICSR lifecycle. For this project, each case was captured in a structured Excel-based dataset at the time of receipt. During intake, available information related to the reporter, patient, suspect drug, and adverse event was recorded without modification or assumption.

The following activities were performed during case intake:

- Assignment of a unique case identifier for traceability
- Documentation of reporter type (e.g., healthcare professional or consumer)
- Recording of patient demographic details as available
- Entry of reported suspect drug(s) and adverse event(s)
- Capture of key dates, including event onset date and case receipt date

If any information was missing at the time of intake, it was documented as Not Available in accordance with pharmacovigilance documentation practices.

B) Case Validity Assessment

Following intake, each case was assessed for minimum validity based on the presence of the four mandatory elements required for a valid ICSR:

1. An identifiable patient
2. An identifiable reporter
3. A suspect drug
4. A reported adverse event

Cases containing all four elements were classified as valid, while cases missing one or more elements were classified as invalid at the time of assessment. The rationale for invalid classification was documented using predefined reasons (e.g., suspect drug not reported, adverse event not reported).

No assumptions were made to complete missing elements, and case validity was determined strictly based on documented information.

C) Seriousness Evaluation

For valid cases, seriousness was evaluated based on the reported adverse event and associated clinical details. Seriousness classification was performed using standard seriousness criteria, including:

- Death
- Life-threatening event
- Hospitalisation or prolongation of hospitalisation
- Disability or permanent impairment
- Congenital anomaly
- Other medically important condition

Cases meeting at least one seriousness criterion were classified as serious, and the applicable seriousness criterion was documented. Cases not meeting any seriousness criteria were classified as non-serious. Seriousness was assessed solely based on reported information without clinical interpretation beyond documentation.

D) Regulatory Timeline Consideration

Regulatory timelines were reviewed to assess compliance awareness as part of the project. Event onset date and case receipt date were used to determine the expected regulatory reporting timeline.

For serious cases, an expedited reporting timeline was considered, while non-serious cases were assessed against standard reporting timelines. Regulatory due dates were calculated and documented to demonstrate understanding of case prioritisation and time-bound processing.

This activity was conducted for training purposes only and did not involve actual regulatory submission.

Summary

This methodology demonstrates a structured, compliant approach to ICSR intake, validation, seriousness assessment, and timeline awareness consistent with Safety Associate responsibilities.

5. Case Narratives

A) Narrative Preparation Approach

As part of this project, 5 Individual Case Safety Report (ICSR) narratives were drafted to demonstrate the ability to convert structured safety data into clear, medically coherent narratives. These narratives were prepared using a standard pharmacovigilance narrative format and were based strictly on the information documented in the dataset.

Each narrative includes:

- Patient demographics (as reported)
- Reporter type and country
- Suspect drug details, including dose and duration where available
- Description of the adverse event
- Seriousness classification and seriousness criteria (if applicable)
- Action taken with the suspect drug
- Concomitant medications and relevant medical history
- Outcome of the event

No assumptions or external clinical interpretations were introduced during narrative preparation. Where information was missing or not reported, this was explicitly stated in the narrative.

Following are the Narratives:

1. Case ID: ICSR-M01
Case Classification: Serious - Recovering
Reporter Type: Healthcare Professional
Country: India

A 45-year-old male patient was reported by a healthcare professional from India. The patient was treated with diclofenac at a dose of 50 mg twice daily for pain. The duration of therapy was 5 days.

On 1-Jan-2025, the patient developed gastrointestinal haemorrhage. The event was considered serious; seriousness was due to hospitalisation. The suspect drug was withdrawn.

Concomitant medications reported included Pantoprazole. The patient's relevant medical history included history of gastritis.

The outcome was reported as recovering.

This case narrative has been prepared for academic training purpose only using a simulated dataset.

2. Case ID: ICSR-M17

Case Classification: Serious – Recovered

Reporter Type: Consumer

Country: India

A 46-year-old male patient was reported by a consumer from India. The patient was treated with glimepiride at a dose of 12 units subcutaneous for diabetes. The duration of therapy was 2 days.

On 1-Feb-2025, the patient developed hypoglycaemia. The event was considered serious; seriousness was due to hospitalisation. The suspect drug was withdrawn.

Concomitant medications reported included metformin. The patient's relevant medical history included diabetes mellitus.

The outcome was reported as recovering.

This case narrative has been prepared for academic training purpose only using a simulated dataset.

3. Case ID: ICSR-M05

Case Classification: Serious – Fatal

Reporter Type: Healthcare Professional

Country: UK

A 70-year-old male patient was reported by a healthcare professional from UK. The patient was treated with warferin at a dose of 75 mg once daily for thrombosis. The duration of therapy was 1 day.

On 11-Jan-2025, the patient developed intracranial haemorrhage. The event was considered serious; seriousness was due to death. The suspect drug was withdrawn.

Concomitant medications reported included aspirin. The patient's relevant medical history included hypertension.

The outcome was reported as fatal.

This case narrative has been prepared for academic training purpose only using a simulated dataset

4. Case ID: ICSR-M27

Case Classification: Non-Serious – Not reported

Reporter Type: Consumer

Country: India

A 41-year-old female patient was reported by a consumer from India. The patient was treated with levothyroxine at a dose of 20 mg once daily for hypothyroidism. The duration of therapy was 3 days.

On 15-Feb-2025, the patient developed palpitations. The event was considered non-serious. The suspect drug was dose reduced.

Concomitant medications reported none. The patient's relevant medical history included palpitations.

The outcome was reported as recovering.

This case narrative has been prepared for academic training purpose only using a simulated dataset.

5. Case ID: ICSR-M29

Case Classification: Non-Serious – Not reported

Reporter Type: Consumer

Country: India

A 26-year-old male patient was reported by a consumer from India. The patient was treated with oral contraceptive at a dose of 500 mg twice daily for contraceptions. The duration of therapy was 2 days.

On 18-Feb-2025, the patient developed nausea. The event was considered non-serious. The suspect drug was continued.

Concomitant medications reported none. The patient's relevant medical history included none.

The outcome was reported as recovered.

This case narrative has been prepared for academic training purpose only using a simulated dataset.

B) Handling of Remaining Cases

The remaining 25 cases were documented and reviewed at a tabular level within the Excel dataset. These cases were assessed for:

- Minimum case validity
- Seriousness classification
- Completeness of core safety data fields

Narrative writing was limited to five cases to maintain focus on quality and structure rather than volume, which is consistent with training-level pharmacovigilance exercises.

Summary

Five cases were selected for detailed narrative writing, while the remaining cases were processed and documented in structured tabular format.

6. Quality Control Checklist

A) Purpose of Quality Control

Basic quality control (QC) checks were performed to demonstrate awareness of data quality, consistency, and documentation standards expected in routine pharmacovigilance operations. These checks were limited to operational review and did not include medical or regulatory validation.

B) Data Completeness Checks

Each case was reviewed to confirm the presence or absence of key safety data elements, including:

- Patient identifiers
- Reporter information
- Suspect drug
- Adverse event
- Key dates

Missing information was not artificially completed and was clearly documented as Not Reported or Unknown, as appropriate.

C) Consistency Checks

Consistency checks were performed to identify:

- Logical alignment between seriousness and seriousness criteria
- Compatibility between adverse events and outcomes
- Coherence between action taken and reported outcome

Any inconsistencies identified were noted and retained as part of the dataset to reflect real-world data variability.

D) Handling of Missing Information

Missing or unavailable information was handled using standard documentation terminology, such as:

- Not Reported
- Unknown
- Not Available

This approach ensures transparency and aligns with accepted pharmacovigilance documentation practices.

Summary

Quality checks focused on data completeness, internal consistency, and appropriate handling of missing information.

7. Summary Analysis & Observations

A high-level operational summary of the processed dataset was performed to demonstrate basic analytical awareness without performing clinical interpretation or signal detection.

Key observations include:

- The dataset comprised a mix of serious and non-serious cases
- Both valid and invalid cases were identified based on minimum case validity criteria
- Gastrointestinal, metabolic, and neurological adverse events were commonly reported
- Most cases contained sufficient information for basic processing, while selected cases demonstrated incomplete reporting
- Regulatory timelines were documented to reflect prioritisation awareness

This analysis was descriptive in nature and intended solely to summarise case characteristics at an operational level.

Summary

The summary analysis provides a descriptive overview of case distribution, data completeness, and processing outcomes.

8. Conclusion & Learning Outcomes

A) Conclusion

This project successfully demonstrates the fundamental pharmacovigilance activities involved in Individual Case Safety Report processing within a simulated environment. Through structured documentation, validity assessment, narrative preparation, and quality checks, the project reflects key responsibilities of an entry-level Safety Associate.

B) Learning Outcomes

Through completion of this project, the following learning outcomes were achieved:

- Practical understanding of minimum ICSR validity requirements
- Improved familiarity with core pharmacovigilance data fields
- Ability to draft structured and compliant case narratives
- Awareness of seriousness classification and regulatory timelines
- Understanding of quality and documentation standards in safety data handling

Alignment with Safety Associate Role

The activities performed in this project align closely with the operational expectations of a Safety Associate, including accurate data entry, structured review, documentation discipline, and quality awareness within pharmacovigilance workflows.

Final Closing Statement

This project serves as a structured demonstration of entry-level pharmacovigilance competencies relevant to a Safety Associate role.

Disclaimer: This document is intended solely for academic and training purposes. All case data are simulated and do not represent real patients or real adverse event reports.